

Sandipan Nath

Koldestraße 29, 91052, Erlangen, Germany
(+49) 15510541744 · sandipan.nath@fau.de
[linkedin.com/in/sandipan-nath](https://www.linkedin.com/in/sandipan-nath) · <https://snath47.github.io/>
Nationality: Indian · **Work Permit:** Germany

ABOUT ME

Data Scientist with a Bachelor of Science in Computing Science and hands-on experience in automation, reinforcement learning, and data systems. Currently pursuing a Master of Science in Data Science at FAU Erlangen-Nürnberg while working at Siemens Energy to optimize HVDC design tools through Python and MATLAB. Specializing in machine learning, neural networks, reservoir computing, reinforcement learning, and scalable data solutions.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

Master of Science in Data Science

Erlangen, Germany

01/04/2024 – Current

- Current Grade: 1.3
- Scale: Best 1.0, Worst 5.0

University of Alberta

Bachelor of Science with Specialization in Computing Science

Edmonton, Canada

04/09/2018 – 22/11/2022

- Final Grade: 3.34
- Scale: Best 4.0, Worst 0.0

WORK EXPERIENCE

Siemens Energy Global A.G.

Automation and Data Science Developer (Working Student)

Erlangen, Germany

07/01/2025 – Current

- Designed and implemented a centralized data management system to streamline data flow between various HVDC system design tools.
- Built a consistent nomenclature system and a supporting database from scratch using Python and MATLAB.
- Developed an interactive dashboard UI to manage tool interoperability via the database, incorporating user feedback to ensure usability.
- Department: HVDC System Design
- 15 hours/week

GaoTek Inc.

Software Development Intern

Toronto, Canada (Remote)

01/05/2023 – 01/09/2023

- Participated in all stages of software development, including design, coding, and testing.
- Created, implemented, and ran automation scripts using open-source tools.
- Collaborated closely with product and design teams using AGILE methodology.
- 20 hours/week

University of Alberta

University Teaching Assistant

Edmonton, Canada

04/01/2021 – 20/04/2021

- Supervised practical labs, marked assignments/tests, and provided coursework assistance.
- 8 hours/week

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, C++, C, Bash/Linux

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Brian, NEST, NESTML, Gymnasium, OpenCV, NLTK, Reinforcement Learning, Reservoir Computing, CNNs, Neural Networks

High-Performance Computing: HPC Clusters, SLURM, Parallel Computing (Multiprocessing/Job Scheduling)

Data & Databases: Pandas, NumPy, MySQL, SQLite, MongoDB, Firebase

Tools & Platforms: Git, Jupyter Notebook, Azure, Matplotlib, Agile SCRUM

SELECTED PROJECTS

- Reinforcement Learning for Control Theory** | *Fraunhofer IIS* Ongoing
- **Key Skills:** Python, Reinforcement Learning, LSTM, Mixture Density Networks (MDN), Control Theory
 - Building a world model using LSTM + MDN to capture and predict complex system dynamics.
 - Training a controller to balance a ball on an arc (on a cart) based on the learned dynamics.
- Forecastability in Chaotic Neural Dynamics** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** Reservoir Computing, Echo State Networks (ESN), Nonlinear Dynamics
 - Built an ESN to model Hindmarsh–Rose neurons in spiking, periodic bursting, and chaotic states.
 - Explored the network’s ability to forecast nonlinear and chaotic time series effectively.
 - Link: github.com/ND-SoSe24-Project/ND-Project
- Object Detection Models for Arctic Wildlife** | *University of Alberta* Completed
- **Key Skills:** Python, Computer Vision, PyTorch
 - Implemented state-of-the-art computer vision models like YOLO for the localization of caribou in camera traps.
 - Evaluated performance in Arctic conditions, focusing on accuracy and computational efficiency.
 - Link: github.com/Robertboy18/Object-Detection-Arctic-Wildlife
- Modelling Drug Effects in Tumor Cell Dynamics** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** ODE Modeling, Systems Biology, SciPy, Numerical Methods
 - Fitted ODE models to describe cell growth, drug availability, and cell killing mechanisms.
 - Simulated and predicted tumor growth responses to various drug regimens.
 - Link: github.com/snath47/oncosys-M1
- Shortest Path in a Train Network** | *FAU Erlangen-Nürnberg* Completed
- **Key Skills:** Dijkstra’s Algorithm, Graph Theory, Data Structures, Python
 - Implemented Dijkstra’s algorithm to determine optimal routes using variable cost functions.
 - Adapted the system to handle multiple constraints such as time, distance, and transfer frequency.
 - Link: github.com/snath47/PathfindingIndianRail

HONOURS AND AWARDS

Murray Thomas Gibson Memorial Scholarship in Mathematics (University of Alberta) – 2019
University of Alberta Undergraduate Scholarship – 2019
Department of Mathematical Statistical Sciences Scholarship (University of Alberta) – 2018

LANGUAGE SKILLS

English: C2 (Proficient/Native level)
Hindi: C2 (Proficient/Native level)
Bengali: Native
German: A2 (Basic)

CERTIFICATIONS

GRE General Test (ETS):
Verbal Reasoning: 161, Quantitative Reasoning: 162, Analytical Writing: 3.5
Scales: Verbal Reasoning: 130-170, Quantitative Reasoning: 130-170, Analytical Writing: 0-6 Mar 2024

TOEFL iBT (ETS):
Total Score 109/120 – Reading: 28/30, Listening: 30/30, Speaking: 23/30, Writing: 28/30 Jan 2024

REFEREES

Prof. Dr. Marius Yamakou
Interim Full Professor (W3)
Department of Data Science (DDS)
Friedrich-Alexander-Universität
Email: marius.yamakou@fau.de

Dr.-Ing. Tobias Feigl
Postdoctoral Scientist
Precise Localization and Data Analytics
Fraunhofer IIS
Email: tobias.feigl@iis.fraunhofer.de

Prof. Dr. Julio Vera-González
Head of "Laboratory of Systems Tumor Immunology"
Hautklinik
Uniklinikum Erlangen
Email: julio.vera-gonzalez@uk-erlangen.de